Pro Football Sports Science Research Update

October 2024

Enhancing team sports performance: A case study of the optimal post-warm-up time window. Sport Sciences for Health, Ahead of Print: 1-10, 2024.

Epidemiological principles in claims of causality: An enquiry into repetitive head impacts (RHI) and chronic traumatic encephalopathy (CTE). Sports Medicine, Ahead of Print: 1-20, 2024.

Estimating muscle fiber-type composition in elite athletes: A survey on current practices and perceived merit. International Journal of Sports Physiology and Performance, Ahead of Print: 1-12, 2024.

Fibrosis and Sports Injuries: Concept and Implications – The GESMUTE-SETRADE Consensus Statement. The Orthopaedic Journal of Sports Medicine, 12(9): 1-4, 2024.

Hamstring muscle-tendon geometric adaptations to resistance training using the hip extension and Nordic hamstring exercises. Scandinavian Journal of Medicine and Science in Sports, 34: e14728, 2024.

Heart rate variability activity in soccer athletes after a musculoskeletal injury. Journal of Rehabilitation Medicine, 56: 1-5, 2024.

Higher levels of morning and daytime light exposure associated with positive sleep indices in professional team sport athletes. Nature and Science of Sleep, 16: 1279-1290, 2024.

Injectable therapeutic peptides – An adjunct to regenerative medicine and sports performance? Arthroscopy: The Journal of Arthroscopic and Related Surgery, Ahead of Print: 1-9, 2024.

Injury-inciting circumstances of sudden-onset hamstring injuries: Video analysis of 63 match injuries in male professional football players in the Qatar Stars League (2013-2020). British Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Meldonium supplementation in professional athletes: Career destroyer or lifesaver? Cureus, 16(7): e63634, 2024.

Multiligament knee injury (MLKI): An expert consensus statement on nomenclature, diagnosis, treatment and rehabilitation. British Journal of Sports Medicine, Ahead of Print: 1-16, 2024.

Nutritional approaches applied to recovery of skeletal muscle injury immobilization: A review of nutrition aid for sport trauma. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-10, 2024.

PEP: A tackle value measuring the prevention of expected points. arXiv:2407.08508v1, 2024.

People are more variable than their hop test would suggest: Hop performance and self-reported outcomes over 11 years following ACL reconstruction. Scandinavian Journal of Medicine and Science in Sports, 34: e14727, 2024.

Professional football (soccer) training and recovery: A longitudinal study on the effects of weekly conditioning session and workload variables. PLOS ONE, 19(9): e0310036, 2024.

Public interest in American on cardiac arrest following cardiovascular events of Bronny and Damar: A Google trend study. American Heart Journal Plus: Cardiology Research and Practice, Ahead of Print: 1-12, 2024.

Repetitive head-injury exposure and later-in-life cognitive and emotional outcomes among former collegiate football players: A CLEAATS investigation. International Review of Psychiatry, 36(3): 233-242, 2024.

Subclinical brain manifestations of repeated mild traumatic brain injury are changed by chronic exposure to sleep loss, caffeine, and sleep aids. Experimental Neurology, 381: 114928, 2024.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Testing limb symmetry and asymmetry after anterior cruciate ligament injury: 4 considerations to increase its utility. Strength and Conditioning Journal, 46(4): 406-413, 2024.

The Braking Performance Framework: Practical recommendations and guidelines to enhance horizontal deceleration ability in multi-directional sports. International Journal of Strength and Conditioning, 4(1): 1-31, 2024.

The effect of a synthetic-grass sport surface on physiology and perception during intermittent exercise in hot conditions. International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2024.

The impact of intrinsic foot muscle strengthening exercises in asymptomatic athletes with normal foot structure: A systematic review. Clinical Epidemiology and Global Health, 27: 101597, 2024.

Training-load management ambiguities and weak logic: Creating potential consequences in sport training and performance. International Journal of Sports Physiology and Performance, Ahead of Print: 1-4, 2024.

Utility of body weight, urine color, and thirst perception (WUT) in determining hydration in young adults. Medicine and Science in Sports and Exercise, Ahead of Print: 1-35, 2024.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Enhancing team sports performance: A case study of the optimal post-warm-up time window. Sport Sciences for Health, Ahead of Print: 1-10, 2024.

Abstract - Warm-up routines are crucial for performance enhancement in team sports, with their effectiveness varying based on individual profiles, exercise mode and timing relative to the activity. This study investigates the impact of a 5–10 min interval post-warm-up on physical performance and sympathetic stimulus, as indicated by salivary biomarkers, in 19 college athletes. Performance was assessed via countermovement jump, 20 m sprint and handgrip strength tests conducted pre-warm-up, 5 min post-warm-up and 10 min post-warm-up. Saliva samples were collected at each stage to evaluate flux, α-amylase activity and protein content. Significant effects ($p < .001$) were observed in all three performance measures and saliva markers. Notably, sprint performance showed the greatest enhancement at the 5 min mark post-warm-up, while jumping improvements persisted through the 10 min window. However, a slight decline in sprint ability was observed from 5 to 10 min post-warm-up. These findings underscore the importance of optimizing warm-up timing and eventually considering re-warm-up strategies to maximize acute performance in team sports.

Epidemiological principles in claims of causality: An enquiry into repetitive head impacts (RHI) and chronic traumatic encephalopathy (CTE). Sports Medicine, Ahead of Print: 1-20, 2024.

Abstract - Determining whether repetitive head impacts (RHI) cause the development of chronic traumatic encephalopathy (CTE)-neuropathological change (NC) and whether pathological changes cause clinical syndromes are topics of considerable interest to the global sports medicine community. In 2022, an article was published that used the Bradford Hill criteria to evaluate the claim that RHI cause CTE. The publication garnered international media attention and has since been promoted as definitive proof that causality has been established. Our counterpoint presents an appraisal of the published article in terms of the claims made and the scientific literature used in developing those claims. We conclude that the evidence provided does not justify the causal claims. We discuss how causes are conceptualized in modern epidemiology and highlight shortcomings in the current definitions and measurement of exposures (RHI) and outcomes (CTE). We address the Bradford Hill arguments that are

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

used as evidence in the original review and conclude that assertions of causality having been established are premature. Members of the scientific community must be cautious of making causal claims until the proposed exposures and outcomes are well defined and consistently measured, and findings from appropriately designed studies have been published. Evaluating and reflecting on the quality of research is a crucial step in providing accurate evidence-based information to the public.

Estimating muscle fiber-type composition in elite athletes: A survey on current practices and perceived merit. International Journal of Sports Physiology and Performance, Ahead of Print: 1-12, 2024.

Purpose: To gather information on practices and perceptions of high-performance experts regarding their athletes' muscle fiber type composition (MFTC) and its estimation.

Methods: A questionnaire on the noninvasive versus invasive estimation of MFTC was completed by 446experts including coaches and sport-science/sports-medicine staff. Moreover, the perceived importance of MFTC for training and performance optimization was assessed. Differences between sport types (individual and team sports) were analyzed using chi-square tests.

Results: Forty percent of the experts implemented MFTC assessment in pursuit of performance optimization, while 50% did not know their athletes' MFTC but expressed a desire to implement it if they would be able to assess MFTC. Ten percent did not perceive value in MFTC assessment. Only 18% of experts believed that their athletes would undergo a muscle biopsy, leading to the adoption of alternative noninvasive techniques. Experts primarily relied on their experience to estimate MFTC (65%), with experts working in individual sports using their experience more frequently than those working in team sports (68% vs 51%; P=.009). Jump tests emerged as the second-most commonly employed method for estimating MFTC (56%). When only considering experts who are currently using MFTC, 87% use MFTC to individualize training volume and 84% to individualize training intensity.

Conclusions: Experts value MFTC assessment primarily to individualize training but mainly rely on noninvasive methods to estimate MFTC. Some of these methods lack scientific validity, suggesting a continuing need for education and further research in this area.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Fibrosis and Sports Injuries: Concept and Implications – The GESMUTE-SETRADE Consensus Statement. The Orthopaedic Journal of Sports Medicine, 12(9): 1-4, 2024.

Background: Musculoskeletal fibrosis can be frequently observed in the scope of sports medicine; however, there is no consensus regarding its definition, nor do we have substantial knowledge regarding its epidemiology or best therapeutic alternatives.

Purpose: The GESMUTE (Group for the Study of the Muscle Tendon System) Epidemiology Group, integrated into SETRADE (The Spanish Society for Sports Traumatology) propose a definition for musculoskeletal fibrosis within the field of physical exercise.

Study design: Consensus statement.

Methods: A bibliographic review of the existing scientific evidence and consensus was developed by the authors on the definition of fibrosis in the field of sport.

Results: Our working group proposed the definition of fibrosis as an abnormal accumulation of extracellular matrix, usually with regards to an injury complication, showing various clinical findings affecting muscles, tendons, ligaments, articular capsules and nerves which, in turn, mars the recovery process, causing symptoms and finally leading to relapses.

Conclusion: Fibrosis in sport may be considered as an abnormal accumulation of extracellular matrix, usually related to complications of an injury, showing a wide range of clinical symptomatology, affecting muscles, tendons, ligaments, articular capsules, and nerves.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Hamstring muscle-tendon geometric adaptations to resistance training using the hip extension and Nordic hamstring exercises. Scandinavian Journal of Medicine and Science in Sports, 34: e14728, 2024.

Abstract - Targeted resistance training stimulates hamstring muscle hypertrophy, but its effect on tendon-aponeurosis geometry is unknown. This study examined changes in hamstring muscle, free tendon, and aponeurosis geometry following a 10-week Nordic or hip extension exercise intervention. Thirty recreationally active males were randomly allocated (n = 10 per group) to a Nordic, hip extension, or control group. Magnetic resonance imaging of both thighs was acquired pre- and post-intervention. Changes in free tendon and aponeurosis volume for each hamstring muscle, biceps femoris long head (BFlh) aponeurosis interface area and muscle volume-to-interface area ratio were compared between groups. Regional changes in muscle CSA were examined via statistical parametric mapping. The change in semimembranosus free tendon volume was greater for the Nordic than control group (mean difference = 0.06 cm^3, 95% CI = 0.02-0.11 cm^3). No significant between-group differences existed for other hamstring free tendons or aponeuroses. There were no between-group differences in change in BFlh interface area. Change in BFlh muscle volume-to-interface area ratio was greater in the hip extension than Nordic (mean difference = 0.10, 95% CI = 0.007-0.19, p = 0.03) and control (mean difference = 0.12, 95% CI = 0.03-0.22, p = 0.009) groups. Change in muscle CSA following training was greatest in the mid-portion of semitendinosus for both intervention groups, and the mid-portion of BFlh for the hip extension group. There was limited evidence for tendon-aponeurosis hypertrophy after 10 weeks of training with the Nordic or hip extension exercises. For the BFlh, neither intervention altered the interface area although hip extension training stimulated an increase in the muscle volume-to-interface area ratio, which may have implications for localized tissue strains. Alternative muscle-tendon loading strategies appear necessary to stimulate hamstring tendon adaptations.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Heart rate variability activity in soccer athletes after a musculoskeletal injury. Journal of Rehabilitation Medicine, 56: 1-5, 2024.

Objectives: The aim of this study is to analyze the adaptations of the autonomic nervous system after a musculoskeletal injury, obtained by measuring heart rate variability in athletes. It was hypothesized that there is an alteration in heart rate variability after a musculoskeletal injury.

Study design: Cohort study.

Subjects: 15 semi-professional soccer players from three football teams, aged between 21 and 33 (mean age: 29.4 ± 3.31 years), with a recent musculoskeletal injury.

Methods: Heart rate variability was collected using the Polar m200 and the chest strap H10 in two moments: within 72 h after the injury and between 5 and 7 days after full return-to-play.

Results: Results show differences between T1 and T2 ($p \leq 0.05$) in low-frequency power (n.u.) ($p = 0.001$) and high-frequency power (n.u.) ($p = 0.001$), in low-frequency/high-frequency ratio ($p = 0.001$) and in high-frequency power (ms2) ($p = 0.017$) measures. No statistical differences were found in low-frequency power (ms2) ($p = 0.233$). The low frequency power (n.u.) was significantly lower after injury compared with LF power (n.u.) values after full return-to-play. In high-frequency power there was a significant difference between both moments with high values after injury.

Conclusions: The use of heart rate variability therefore seems to be promising to detect an imbalance in the autonomic nervous system and help clinical departments to identify a possible non-traumatic musculoskeletal injury. Further research should be performed considering a wide range of musculoskeletal injuries and to establish baseline values of the athletes.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Higher levels of morning and daytime light exposure associated with positive sleep indices in professional team sport athletes. Nature and Science of Sleep, 16: 1279-1290, 2024.

Objective: Light exposure techniques have been recommended to combat sleep issues caused by disruption to circadian regularity in the athletic population, although studies are lacking.

Methods: A total of 17 professional male Australian Football athletes (age ± SD: 22 ± 3 years) wore a wrist actigraph to measure sleep parameters, and a wearable light sensor to measure melanopic equivalent daylight illuminance (mEDI, in lux) for 14 days. Participants completed three sleep questionnaires at the end of the data collection period and completed well-being surveys 6 times. The Sleep Regularity Index (SRI) for each player was also calculated from actigraphy data. Light exposure data were organized into three different timeframes: morning (wake time + 2 hours), daytime (end of morning to 6 pm), and evening (2 hours leading up to bedtime) for analysis. Repeated measures correlation was conducted for objective sleep measures and mEDI values per timeframe. Pearson's correlation was conducted on subjective sleep measures and well-being measures against mEDI values per timeframe.

Results: Higher morning light was associated with significantly ($p < 0.001$) greater total sleep time ($r = 0.31$). Higher daytime light exposure was associated with higher subjective sleep quality ($r = 0.48$, $p < 0.05$). Higher evening light exposure was associated with higher Athlete Sleep Screening Questionnaire (ASSQ) global scores ($r = 0.52$, $p < 0.05$). There were no other significant correlations between light exposure and sleep or well-being measures ($p > 0.05$).

Conclusion: Higher morning and daylight exposure levels were associated with various positive objective and subjective sleep measures in professional team sport athletes, supporting the need for education on optimizing light exposure to improve circadian function, sleep, and health.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Injectable therapeutic peptides – An adjunct to regenerative medicine and sports performance? Arthroscopy: The Journal of Arthroscopic and Related Surgery, Ahead of Print: 1-9, 2024.

Abstract - High level athletes and bodybuilders are constantly seeking novel therapies to enhance recovery and expedite return from injury-injectable peptides are a new and trending therapy that may be the wave of the future of in the realm of regenerative medicine research in treating joint injuries and osteoarthritis. Very early in vivo research of the pharmacokinetics suggests the possibility of Body Protection Compound 157 (BPC-157) appears to be at the forefront of therapeutic peptides with early demonstration of these experimental peptides optimizing endurance training, metabolism, recovery and tissue repair. While unregulated, yet and readily available for purchase over the internet, there is scarce orthopaedic literature investigating the clinical use and outcomes of these therapeutic peptides in tendon, muscle and cartilage injury. However, this has not slowed the recent exponential growth of the multi-billion-dollar industry in development of therapeutic peptides. As orthopaedic surgeons and team physicians, we should stay up-to-date with the latest pharmacokinetic, safety, ethical, and legal profiles regulations regarding synthetic peptide supplementation for injury recovery and sports performance optimization from our elite athletes to our fitness fanatics as these patients will continue to seek the latest and greatest in treatment options and will be approaching us with questions on their results, risks and benefits.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Injury-inciting circumstances of sudden-onset hamstring injuries: Video analysis of 63 match injuries in male professional football players in the Qatar Stars League (2013-2020). British Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

Objective: To describe and categorize the injury-inciting circumstances of sudden-onset hamstring match injuries in professional football players using systematic video analysis.

Methods: Using a prospective injury surveillance database, all sudden-onset hamstring match injuries in male football players (18 years and older) from the Qatar Stars League between September 2013 and August 2020 were reviewed and cross-referenced with broadcasted match footage. Videos with a clear observable painful event (ie, a player grabbing their posterior thigh) were included. Nine investigators independently analyzed all videos to describe and categorize injury-inciting circumstances. We used three main categories: playing situation (eg, time of injury), player action(s) (eg, running) and other considerations (eg, contact). Player action(s) and other considerations were not mutually exclusive.

Results: We included 63 sudden-onset hamstring match injuries out of 295 registered injuries between 2013 and 2020. Running was involved in 86% of injuries. Hamstring injuries occurred primarily during acceleration of 0-10 m (24% of all injuries) and in general at different running distances (0-50 m) and speeds (slow to fast). At 0-10 m distance, indirect player-to-player contact and inadequate balance were involved in 53% and 67% of the cases, respectively. Pressing occurred in 46% of all injuries (injured player pressing opponent: 25%; being pressed by opponent: 21%) and frequently involved player-to-player contact (69% of the cases when the injured player was pressing vs 15% of the cases when the opponent was pressing) and inadequate balance (82% vs 50%, respectively). Other player actions that did not involve running (n=9, 14% of all injuries) were kicking (n=6) and jumping (n=3).

Conclusion: The injury-inciting circumstances of sudden-onset hamstring match injuries in football varied. The most common single-player action (24%) was acceleration over a distance of <10 m. Pressing, inadequate balance and indirect contact were frequently seen player actions. Injury prevention research in football should look beyond high-speed running as the leading risk factor for sudden-onset hamstring injuries.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Hamstring muscle injury is the most common injury type in football.

⇒ Sudden- onset hamstring injuries are reported to happen during high- speed running or stretching (hip flexion and knee extension).

WHAT THIS STUDY ADDS

⇒ Sudden- onset hamstring injuries happen more frequently at lower speeds and distances than previously thought.

⇒ Pressing, indirect contact and inadequate balance are frequent factors involved in these injury situations.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Research and application of prevention and rehabilitation strategies may need to consider injury mechanisms beyond high- speed running, especially short accelerations.

⇒ The complexity of movements inherent in sport such as football is highlighted as a potential risk factor for injury.

⇒ Introducing sport- specific elements, such as playing situations, indirect contact and unexpected changes to demands for performance may benefit injury prevention programs in the future.

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Meldonium supplementation in professional athletes: Career destroyer or lifesaver? Cureus, 16(7): e63634, 2024.

Abstract - Meldonium is a substance with known anti-anginal effects demonstrated by numerous studies and human clinical trials; however, it does not possess marketing authorization within the European Union, only in ex-Soviet republics. Since 2016, meldonium has been included by the World Anti-doping Agency (WADA) on the S4 list of metabolic modulators. In performance athletes, meldonium is now considered a doping agent due to its capacity to decrease lactate production during and after exercise, its capability to enhance the storage and utilization of glycogen, and its protective action against oxidative stress. Together, these attributes can significantly improve aerobic endurance, cardiac function, and capacity as well as shorten recovery times (allowing higher intensity training), thereby enhancing performance. The purpose of this review is to highlight the most important mechanisms underlying the protective effect of meldonium against mitochondrial dysfunction (MD), which is responsible for oxidative stress, inflammation, and the cardiac changes known as "athletic heart syndrome." Meldonium acts as an inhibitor of γ-butyrobetaine hydroxylase (BBOX), preventing the de novo synthesis of carnitine and its absorption at the intestinal level via the organic cation/carnitine transporter 2 (OCTN2) and directing the oxidation of fatty acids to the peroxisomes. The decrease in mitochondrial β-oxidation of fatty acids leads to a reduction in lipid peroxidation products that cause oxidative stress and prevent the formation of acyl/acetyl-carnitines involved in numerous pathological disorders. Given the recent findings of the potentially detrimental effects of prolonged high-intensity exercise on cardiovascular health and coronary atherosclerosis, there may be legitimate arguments for the justification of the use of substances like meldonium as protective supplements for athletes.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Multiligament knee injury (MLKI): An expert consensus statement on nomenclature, diagnosis, treatment and rehabilitation. British Journal of Sports Medicine, Ahead of Print: 1-16, 2024.

Abstract - Multiligament knee injuries (MLKIs) represent a broad spectrum of pathology with potentially devastating consequences. Currently, disagreement in the terminology, diagnosis and treatment of these injuries limits clinical care and research. This study aimed to develop consensus on the nomenclature, diagnosis, treatment and rehabilitation strategies for patients with MLKI, while identifying important research priorities for further study. An international consensus process was conducted using validated Delphi methodology in line with *British Journal of Sports Medicine* guidelines. A multidisciplinary panel of 39 members from 14 countries, completed 3 rounds of online surveys exploring aspects of nomenclature, diagnosis, treatment, rehabilitation and future research priorities. Levels of agreement (LoA) with each statement were rated anonymously on a 5-point Likert scale, with experts encouraged to suggest modifications or additional statements. LoA for consensus in the final round were defined 'a priori' if >75% of respondents agreed and fewer than 10% disagreed, and dissenting viewpoints were recorded and discussed. After three Delphi rounds, 50 items (92.6%) reached consensus. Key statements that reached consensus within nomenclature included a clear definition for MLKI (LoA 97.4%) and the need for an updated MLKI classification system that classifies injury mechanism, extent of non-ligamentous structures injured and the presence or absence of dislocation. Within diagnosis, consensus was reached that there should be a low threshold for assessment with CT angiography for MLKI within a high-energy context and for certain injury patterns including bicruciate and PLC injuries (LoA 89.7%). The value of stress radiography or intraoperative fluoroscopy also reached consensus (LoA 89.7%). Within treatment, it was generally agreed that existing literature generally favors operative management of MLKI, particularly for young patients (LoA 100%), and that single-stage surgery should be performed whenever possible (LoA 92.3%). This consensus statement will facilitate clinical communication in MLKI, the care of these patients and future research within MLKI.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Nutritional approaches applied to recovery of skeletal muscle injury immobilization: A review of nutrition aid for sport trauma. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-10, 2024.

Abstract - Skeletal muscle (SM) injuries are a very common occurrence among athletes during training and/or competitive periods, resulting in time loss and absence from sports. Of several existing interventions to improve SM recovery, the nutritional approach has emerged as integral part to improve the physiological response and maintain the body composition to promote a rapid and safe return to play. Growing evidence supports that the physiological response to SM injuries results in a number of changes, suggesting the relevance of a personalized nutritional approach, according to the degree of immobilization, SM mass loss, strength and function. Therefore, we aimed to review the current evidence of the nutritional approach applied to SM injuries recovery in athletes, highlighting energy intake, use the main nutrients and elements for the SM recovery, such as proteins, carbohydrates, vitamins and omega-3 fatty acids. Currently, evidence from injured athletes is scarce, and so specific guidelines cannot be established, calling from greater research into nutritional interventions in SM injury recovery in athletes.

PEP: A tackle value measuring the prevention of expected points. arXiv:2407.08508v1, 2024.

Abstract - Traditional assessments of tackling in American Football often only consider the num ber of tackles made, without adequately accounting for their context and importance for the game. Aiming for improvement, we develop a metric that quantifies the value of a tackle in terms of the prevented expected points (PEP). Specifically, we compare the real end-of-play yard line of tackles with the predicted yard line given the hypo thetical situation that the tackle had been missed. For this, we use high-resolution tracking data, that capture the position and velocity of players, and a random forest to account for uncertainty and multi-modality in yard-line prediction. Moreover, we acknowledge the difference in the importance of tackles by assigning an expected points value to each individual tree prediction of the random forest. Finally, to re late the value of tackles to a player's ability to tackle, we fit a suitable mixed-effect model to the PEP values. Our approach contributes to a deeper understanding of defensive performances in American football and offers valuable insights for coaches and analysts.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

People are more variable than their hop test would suggest: Hop performance and self-reported outcomes over 11 years following ACL reconstruction. Scandinavian Journal of Medicine and Science in Sports, 34: e14727, 2024.

Abstract - We aimed to report the trajectory of self-reported outcomes up to 11 years post-ACLR. We also explored the relationship between hop performance at 1 year and: (i) future self-reported knee outcomes; and (ii) risk of subsequent knee events. 124 participants (43 women, mean age 31 ± 8 years) were recruited at 1 year following hamstring-autograft ACLR. Hop performance was assessed with single-forward and side-hop tests. Follow-up was completed at 3 (n = 114), 5 (n = 89) and 11 years (n = 72) post-ACLR. Self-reported outcomes were assessed at each follow-up with the Knee injury Osteoarthritis Outcome Score (KOOS) pain and quality of life (QOL) subscales. Generalized linear mixed models estimated the relationship between hop performance and self-reported outcomes. Subsequent knee events (new injury/surgery) to either knee were recorded, with the relationship between hop performance and risk of subsequent knee events analyzed with Cox proportional hazards. Self-reported knee outcomes were stable (mean change < 10 points) across all timepoints but with major within-sample variability. There was a modest relationship between greater hop performance at 1 year and better future KOOS-pain (average marginal effect [AME] % improvement with + 1 cm single forward hop = 0.06% [95% CI 0.02-0.10]). A nonlinear spline relationship showed better single-forward hop performance was associated with better KOOS-QOL for scores < 108 cm, not present for higher hop scores > 108 cm. There were 21 index and 11 contralateral subsequent knee events. Hop performance was not related to risk of a subsequent knee event (hazard ratio index knee 0.99 [95% CI 0.98-1.02]). In conclusion, self-reported knee pain and quality of life were generally stable across the 11-year follow-up period. Greater hop performance at 1-year post-ACLR was related to better self-reported knee outcomes up to 11-year follow-up (of questionable clinical importance), but not associated with the risk of subsequent knee injury/surgery.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Professional football (soccer) training and recovery: A longitudinal study on the effects of weekly conditioning session and workload variables. PLOS ONE, 19(9): e0310036, 2024.

Abstract - The main purpose of this longitudinal study was to investigate football players' recovery status, through hormonal response, in relation to accumulated workload at two comparable time points of the first (T1) and second half (T2) of the competitive season. Moreover, this study investigated athletes' hormonal response to a typical weekly conditioning session (5 days before match: MD-5), at T1 and T2, to detect changes in players' recovery capability over time. Salivary cortisol (sC) and testosterone (sT) of 24 professional players (27.8 ± 4.1 years of age) were collected before, after, and 24 hours following MD-5 in two comparable microcycles of T1 and T2. GPS training data (total and high-intensity distance) of the 7 and 28 days before sampling were used to obtain athletes' acute and chronic workloads. Results showed a pre-training significant decrease of sT and an increase of sC ($p<0.05$) in T2, compared to T1. Moreover, athletes showed high sC and low sT levels before, after and 24 hours following MD-5 in T2. Workload analysis revealed significant correlations of chronic load with sC ($r = 0.45$, $p = 0.056$) and T/C ratio ($r = -0.59$; $p = 0.007$). These results suggested that, in professional football, chronic workload has a greater impact on players' recovery time than acute workload over the sport season. Moreover, athletes' hormonal response to the weekly conditioning session at T2 revealed an altered anabolic/catabolic balance, highlighting the key role of continuous internal and external workload monitoring during the season.

Public interest in American on cardiac arrest following cardiovascular events of Bronny and Damar: A Google trend study. American Heart Journal Plus: Cardiology Research and Practice, Ahead of Print: 1-12, 2024.

Background: Heart disease is one of the leading causes of death in the United States. Increased education and utilization of BLS by first responders have had a significant impact, but certain populations remain high risk, such as African Americans. Raising awareness among at-risk populations may lead to more bystander CPR performed, improving mortality rates. The influence of celebrity deaths

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

and illnesses is an important driver of public awareness. Therefore, the cardiac arrests of both Bronny James and Damar Hamlin may have influenced cardiac arrest awareness.

Methods: Google Trends data was pulled for the following search terms from 8/21/2022-8/14/2023: Cardiac arrest (disease), Cardiopulmonary Resuscitation (topic), Basic Life Support (topic), Myocardial Infarction (disease), Defibrillation (topic) and Automatic External Defibrillator (topic). The average relative search volume (RSV) for each search term was taken for a three-week period encompassing the week of and two weeks following the cardiac arrests of Damar Hamlin and Lebron James Jr., respectively. We used one-way ANOVA and independent sample t-tests to compare the average values of Damar Hamlin's and LeBron James Jr.'s incidents with their respective 12-month averages.

Results: RSV was significantly higher surrounding Hamlin's cardiac arrest compared to James Jr.'s for Cardiopulmonary Resuscitation and Automatic External Defibrillator. RSV for Basic Life Support was increased in LeBron James Jr.'s time compared to the 12-month average and Damar Hamlin's incident. Compared to the 12-month average, Cardiac arrest, Cardiopulmonary Resuscitation, Defibrillation, and Automatic External Defibrillator during Hamlin's incident. Myocardial infarction RSV was higher during James Jr.'s incident compared to baseline. Over the long term, the search terms showed a significant increase after Damar Hamlin's incident when compared to before.RSV was significantly higher surrounding Hamlin's cardiac arrest compared to James Jr.'s for "Cardiopulmonary Resuscitation" (23.56 vs. 22.0, $p < 0.00$) and "Automatic External Defibrillator" (19.59 vs. 19.4, $p < 0.00$). RSV for "Basic Life Support" was increased in LeBron James Jr.'s time compared to the 12-month average and Damar Hamlin's incident (80.9 vs. 66.88, $p = 0.04$). Compared to the 12-month average, "Cardiac arrest," "Cardiopulmonary Resuscitation," "Defibrillation," and "Automatic External Defibrillator" during Hamlin's incident showed significant increases. "Myocardial infarction" RSV was higher during James Jr.'s incident compared to baseline (55 vs. 46.6, $p = 0.026$). Over the long term, the search terms showed a significant increase after Damar Hamlin's incident when compared to before ($p < 0.05$).

Conclusions: Increases in the search terms for Hamlin's cardiac arrest compared to James Jr.'s cardiac arrest were associated with seeing the event live and increasing cardiac arrest awareness. Hamlins Cardiac Arrest also showed a significant increase in search terms over the long term. The increase in searches for "Basic Life Support" during James Jr.'s cardiac arrest indicates increased awareness. Also, the increase in myocardial infarction searches during both incidents could show confusion between cardiac arrest and myocardial infarction.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Repetitive head-injury exposure and later-in-life cognitive and emotional outcomes among former collegiate football players: A CLEAATS investigation. International Review of Psychiatry, 36(3): 233-242, 2024.

Abstract - This study measured the relationship between head-injury exposure and later-in-life cognitive and emotional symptoms in aging collegiate football players who participated in the College Level Aging Athlete Study. Linear regressions examined the relationship between various head-injury exposure variables (head-injury exposure estimate [HIEE], number of diagnosed concussions, and symptomatic hits to the head) and subjective cognitive function, objective cognitive function, and emotional/mood symptoms. Additional regressions evaluated the impact of emotional symptoms on subjective cognitive decline and objective cognitive function. Participants (n = 216) were 50-87 years old (M = 63.4 [8.5]), 91% White, and well-educated (bachelor's/graduate degree = 92%). HIEE did not predict scores on cognitive or emotional/mood symptom measures (p's > .169). Diagnosed concussions had a small effect on depression symptoms (p = .002, b = 0.501, R2 = .052) and subjective cognitive symptoms (p = .002, b = 0.383, R2 = .051). An emotional symptom index had a stronger relationship (p < .001, b = 0.693, R2 = .362) with subjective cognitive functioning but no significant relationship with objective cognitive function (p = .052, b = -0.211, R2 = .020). Controlling for emotional symptoms, the relationship between concussions and subjective cognitive symptoms was attenuated (p = .078, R2 = .011). Findings suggested that head-injury exposure was not significantly related to cognitive or emotional/mood outcomes in former collegiate football players and highlighted the importance of current emotional/mood symptoms on subjective cognitive function.

Subclinical brain manifestations of repeated mild traumatic brain injury are changed by chronic exposure to sleep loss, caffeine, and sleep aids. Experimental Neurology, 381: 114928, 2024.

Introduction: After mild traumatic brain injury (mTBI), the brain is labile for weeks and months and vulnerable to repeated concussions. During this time, patients are exposed to everyday circumstances that, in themselves, affect brain metabolism and blood flow and neural processing. How commonplace activities interact with the injured brain is unknown. The present study in an animal model investigated

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

the extent to which three commonly experienced exposures-daily caffeine usage, chronic sleep loss, and chronic sleep aid medication-affect the injured brain in the chronic phase.

Methods: Subclinical trauma by repeated mTBIs was produced by our head rotational acceleration injury model, which causes brain injury consistent with the mechanism of concussion in humans. Forty-eight hours after a third mTBI, chronic administrations of caffeine, sleep restriction, or zolpidem (sedative hypnotic) began and were continued for 70 days. On Days 30 and 60 post injury, resting state functional magnetic resonance imaging (fMRI) and diffusion tensor imaging (DTI) were performed.

Results: Chronic caffeine, sleep restriction, and zolpidem each changed the subclinical brain characteristics of mTBI at both 30 and 60 days post injury, detected by different MRI modalities. Each treatment caused microstructural alterations in DTI metrics in the insular cortex and retrosplenial cortex compared with mTBI, but also uniquely affected other gray and white matter regions. Zolpidem administration affected the largest number of individual structures in mTBI at both 30 and 60 days, and not necessarily toward normalization (sham treatment). Chronic sleep restriction changed local functional connectivity at 30 days in diametrical opposition to chronic caffeine ingestion, and both treatment outcomes were different from sham, mTBI-only and zolpidem comparisons. The results indicate that commonly encountered exposures modify subclinical brain activity and structure long after healing is expected to be complete.

Conclusions: Changes in activity and structure detected by fMRI are widely understood to reflect changes in the functions of the affected region which conceivably underlie mTBI neuropathology and symptomatology in the chronic phase after injury.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Testing limb symmetry and asymmetry after anterior cruciate ligament injury: 4 considerations to increase its utility. Strength and Conditioning Journal, 46(4): 406-413, 2024.

Abstract - Anterior cruciate ligament (ACL) injury occurs frequently in sport and surgical reconstruction is often recommended to restore knee joint stability. To guide rehabilitation and determine return to sport readiness, practitioners have used a long-standing practice of calculating the limb symmetry index (LSI) in various functional, biomechanical, and strength tests to compare the injured limb with the non-injured contralateral limb. However, the evidence in support of the LSI calculation to quantify rehabilitation status and return to sport readiness is mixed. We synthesize scientific literature on the LSI calculation and discuss potential reasons for the mixed evidence and limitations. We present 4 considerations to improve the utility of the LSI calculation including (a): the importance of establishing the right benchmark of recovery such as the preinjury contralateral limb or a sport-specific non-injured control benchmark; (b) strategies to manage the high variation in movement asymmetry calculations and the importance of quantifying the intrasubject variability for the component parts of the LSI; (c) the evidence for assessing the movement strategy alongside performance when using the LSI; and (d) how a sport-specific envelope of function can be used to inform post-ACL injury testing that incorporates the LSI.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

The Braking Performance Framework: Practical recommendations and guidelines to enhance horizontal deceleration ability in multi-directional sports. International Journal of Strength and Conditioning, 4(1): 1-31, 2024.

Abstract - Horizontal deceleration is a critical locomotor skill underpinning the many changes of speed and direction performed by athletes in multi-directional sports. Despite its importance there are currently no evidence-based guidelines on how to improve horizontal deceleration ability. Therefore, the aim of this article is to provide practitioners tasked with preparing athletes for the demands of multi-directional sports with practical recommendations and guidelines on how to improve their athlete's horizontal deceleration ability. The article proposes the concept and use of a Braking Performance Framework to help guide the selection of training methods and exercises to target the currently known neuromuscular and biomechanical determinants of horizontal deceleration ability. The Braking Performance Framework provides practitioners with evidence-informed training methods that can be integrated with other important physical, technical, and tactical components to optimize a player's horizontal deceleration ability.

The effect of a synthetic-grass sport surface on physiology and perception during intermittent exercise in hot conditions. International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2024.

Purpose: The current study aimed to determine the effect of a synthetic-grass sport surface on core body temperature, skin temperature, heart rate, thermal sensation, thermal comfort, and rating of perceived exertion (RPE) during intermittent exercise in hot conditions.

Methods: Using a randomized crossover design, 13 trained/developmental team-sport athletes completed two 50-minute standardized intermittent running protocols on a synthetic and a natural-grass surface, on separate days (control-condition air temperature 32.6 °C [1.3 °C], relative humidity 43.2% [5.3%]).

Results: Final skin temperature was significantly higher on synthetic compared with natural grass at the calf (40.1 °C [2.5 °C] vs 33.4 °C [0.6 °C]; P < .001), shoulder (36.6 °C [1.7 °C] vs 33.7 °C [0.7 °C]; P < .001),

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

and chest (33.2 °C [1.1 °C] vs 31.8 °C [1.2 °C]; P = .02). Thermal sensation (median: 2.3; interquartile range [0.5] vs 2.2 [0.5], P = .03) and sweat rate (1.5 [0.4] L·h-1 vs 1.2 [0.3] L·h-1; P = .02) were also significantly higher on synthetic grass. While final core body temperature was significantly higher on the natural than synthetic grass (38.4 °C [0.3 °C] vs 38.2 °C [0.4 °C]), there were no significant differences in delta core temperature, as well as heart rate, thermal comfort, or RPE.

Conclusions: Higher skin temperatures, thermal sensation, and sweat rates suggest that exercising on synthetic grass in hot conditions may increase some markers of heat strain during exercise. However, delta core body temperature, heart rate, thermal comfort, and RPE remained unaffected.

The impact of intrinsic foot muscle strengthening exercises in asymptomatic athletes with normal foot structure: A systematic review. Clinical Epidemiology and Global Health, 27: 101597, 2024.

Purpose: The intrinsic foot muscles (IFM) are crucial to the passive, active, and neurological subsystems that compose the foot core, and their weakness may predispose to various foot pathologies, systemic diseases and musculoskeletal conditions. Strengthening of IFM is an effective intervention in clinical conditions, but its role in sports and clinical rehabilitation is still inconclusive. Thus, this review aims to evaluate the effects of IFM strengthening exercises in athletes with normal feet.

Methods: An electronic database search was conducted using Cochrane Central Register of Controlled Trials (CENTRAL), Web of Science (WoS), Scopus, MEDLINE, PubMed, and the Physiotherapy Evidence Database (PEDro). The Cochrane risk of bias tool was used for assessing bias in the included studies.

Results: Four studies were included for qualitative review. The meta-analysis was not possible due to the heterogeneity of measuring units of outcome measures in the included studies. All the studies showed an improvement in intrinsic foot muscle strength, though the training protocols used by them were only roughly equivalent. One study reported no significant differences in MLA stiffness after 8 weeks of intervention. The clinician assessed motor performance and morphological assessments showed considerable improvements. Improved vertical running propulsive forces were reported, though the running economy showed no changes.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Conclusion: Findings suggest that IFM exercises effectively improve outcomes such as foot morphology and function, strength, running mechanics and subjective function in athletes and may be incorporated in sports rehabilitation. These conclusions should be considered with caution due to bias in included studies.

Highlights

• Intrinsic Foot Muscle (IFM) exercises improve foot strength and morphology in athletes, supporting injury prevention.

• All the studies showed an improvement in IFM strength.

• The clinician assessed motor performance and morphological assessments showed considerable improvements.

• Improved vertical running propulsive forces were reported, though the running economy showed no changes.

• Findings suggest that IFM exercises may be incorporated in sports rehabilitation.

Training-load management ambiguities and weak logic: Creating potential consequences in sport training and performance. International Journal of Sports Physiology and Performance, Ahead of Print: 1-4, 2024.

Background: The optimization of athlete training load is not a new concept; however in recent years, the concept of "load management" is one of the most widely studied and divisive topics in sports science and medicine.

Purpose: Discuss the challenges faced by sports when utilizing training load monitoring and management, with a specific focus on the use of data to inform load management guidelines and policies/mandates, their consequences, and how we move this field forward.

Challenges: While guidelines can theoretically help protect athletes, overzealous and overcautious guidelines may restrict an athlete's preparedness, negatively influence performance, and increase injury

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

risk. Poor methods, wrong interpretation of study findings, and faulty logic do not allow for systematic scientific evaluations to inform guidelines.

Practical Solutions: Guidelines and mandates should be developed through a systematic research process with stronger research designs and clear research questions. Collaborating with statistical and epidemiological experts is essential. Implementing open science principles and sharing all sports training load data increase transparency and allow for more rapid and valid advancements in knowledge. Practitioners should incorporate multiple data streams and consider individual athlete responses, rather than applying broad guidelines based on average data.

Conclusion: Many current training load guidelines and mandates in sports come from good intentions; however, they are arbitrary without sound knowledge of the underlying scientific principles or methods. Common sense guidelines are helpful when there is sparse literature, but they should be careful to avoid arbitrarily choosing findings from weak research. Without precise scientific inquiries, implementing training load interventions or guidelines can have negative implications.

Utility of body weight, urine color, and thirst perception (WUT) in determining hydration in young adults. Medicine and Science in Sports and Exercise, Ahead of Print: 1-35, 2024.

Objective: The primary aim of this study was to assess the efficacy of the weight, urine, thirst (WUT) framework in predicting dehydration after a body water manipulation protocol, while concurrently determining the individual and interactive contributions of the model components.

Methods: The total study sample was 93 participants (female, n = 47), recruited from two institutions. Phase 1 involved collecting daily hydration measures from free-living participants (Study 1, 58 participants for 3 days; Study 2, 35 participants for 7 days). Phase 2 entailed a two-hour passive heating protocol, where participants from Study 2 were randomly assigned to one of three groups that manipulated total body water over 24-hours using passive heating and fluid restriction. During each Phase, participants provided urine samples, underwent body mass measurements, and completed questionnaires pertaining to thirst perception. Morning and 24-hour urine samples were assessed for

color, osmolality, and specific gravity. Differences between intervention groups, based on the probability of hydration status, were examined (ANOVA) and ridge regression analysis assessed the relative importance of variables within the WUT model.

Results: The study revealed significant differences among the intervention groups for predicted probability of dehydration, as determined by changes in body mass ($p = 0.001$), urine color ($p = 0.044$), and thirst perception ($p < 0.001$). Binomial ridge regression indicated that change in body mass (58%) and thirst perception (26%) were the most influential predictors of dehydration.

Conclusions: These data support use of an enhanced version of the WUT model, underscoring the significance of changes in body mass and thirst perception in the assessment of hydration status.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/